

PULSE Instrument Cluster

Sprint 2 Backlog

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Document control

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Status	Draft for the sprint 1 review; frozen after review feedback is merged
Date	2026-10-02
Author	Ahmed Abdelghany
Reviewer	Cockpit Electronics / HMI Platform Group (sponsor)
Sprint window	Week 4 to week 6, 7 September to 25 September 2026
Review gate	Week 6 demo: signal layer and Flutter cluster verified on live broker data
Related	PULSE-StRS-001, PULSE-SyRS-001, PULSE-SAF-001, PULSE-SEC-001

Change log

Version	Date	Author	Change
0.1	2026-09-03	A. Abdelghany	First draft from the delivery plan and the week-3 status columns; review feedback section left open
0.2	2026-09-08	A. Abdelghany	Document-control fields corrected so the review gate and related documents resolve. Item 7 reduced to Partial: the databroker digest is pinned, the Flutter SDK version is not. Sprint goal and out-of-scope wording clarified
0.3	2026-09-11	A. Abdelghany	Week 4 checkpoint added as section 8: per-item state verified against the repository, hours position, decisions requested. Sprint 1 review recorded as held with no comments
0.4	2026-09-23	A. Abdelghany	Week 6 checkpoint added as section 8.5: state per item on 23 September, definition of done against the gate, decisions taken while implementing
0.5	2026-10-02	A. Abdelghany	Section 8.6: the three partial items closed on a second host, findings, and the week 7 checkpoint of sprint 3

1. Sprint goal

Close work package 02. An automated suite verifies the signal layer and the Flutter cluster against live broker data, and runs before every demo. Every desk-only shortcut that has a sprint 2 requirement attached is removed. The week 6 demo shows the same cluster as week 3, now with contract tests, degraded-state display, encrypted transport and pinned upstream versions behind it.

2. Capacity

Item	Value
Engineers	2 at 40 h a week
Hours in sprint	240
Committed below	196
Reserve for review feedback and risk	44

3. Backlog

Ordered by priority. Every item names the requirement it moves and the status it should reach by the week 6 gate. Estimates are engineering hours including verification.

#	Item	Moves	To status	Est. h	Verification
1	Regression suite in CI shape: broker up, signal contract, VSS regeneration, live telemetry, HMI purity, secrets, Flutter launch. Run recorded under docs/evidence/ before every demo	CS-11, SR-9, FR-11, FR-3	Implemented	16	Suite passes on a clean checkout
2	Signal contract test against the running broker: every path in PULSE-SAD-001 appendix A subscribes and receives a typed datapoint	FR-11, FR-13, IR-2	Implemented	12	Test in the suite

#	Item	Moves	To status	Est. h	Verification
3	UI smoke test per build: cluster launches, connects, renders a known injected value, exits clean; Compose cluster on the emulator	FR-12, FR-14	Implemented	20	Test in the suite
4	Staleness detection in the Flutter cluster: freshness criterion per safety-relevant signal from PULSE-SAF-001 section 3, degraded presentation, broker loss shows all stale	SR-4, SR-5	Implemented	12	Test: kill the broker, dashes appear within the criterion
5	Same staleness handling in the Compose cluster	SR-4, SR-5	Implemented	16	Test on the emulator
6	Mutual TLS on the databroker with clearly marked development certificates; start-up prints a warning whenever --insecure is used; both clients and both providers connect over TLS	CS-3, IR-5	Implemented	24	Suite: plaintext client refused unless the development flag is set
7	Pin the Flutter SDK version and record it in the README; the databroker image digest is already pinned	NFR-8, DD-10, DD-11	Partial	6	Inspection
8	Software bill of materials per component (CycloneDX) and a vulnerability check in the suite	CS-8	Implemented	14	Suite
9	Driver-monitor privacy test: no file written, no socket other than the broker	CS-6	Implemented	6	Suite
10	HMI-side range and enumeration check as defence in depth, with a test that an out-of-range publish is refused by the broker	CS-5	Implemented	10	Suite
11	Catalogue checksum in the suite, so a tampered vss_agl.json fails the run	CS-8, T-4	Implemented	4	Suite
12	Merge the driver-monitoring branch into main after review; one baseline for sprint 2	NFR-5	Done	4	Onboarding guide re-run on the merged tree
13	Software architecture description for the Flutter cluster and the providers (SWAD), one page each, as promised in PULSE-SAD-001 section 1	SR-10	Implemented	16	Review
14	Safety and security review of PULSE-SAF-001 and PULSE-SEC-001 with the assigned managers; ratings confirmed or changed	SR-1, SR-2, CS-1	Reviewed	8	Review minutes
15	Virtual CAN feeder spike: vcan0 with kuksa-can-provider and the AGL DBC, publishing three signals, to de-risk sprint 3	FR-8	Partial	8	Demonstration

4. Out of scope this sprint

Six items stay in sprints 3 and 4, as the delivery plan in PULSE-StRS-001 section 3.1 already schedules them: physical CAN hardware, the Raspberry Pi 5 baseline, per-client write authorisa-

tion (CS-4), security event logging (CS-7), the fail-visible renderer (SR-6) and the Autoware bridge (FR-22).

5. Risks carried into the sprint

Risk	Response
TLS in the Dart and Kotlin gRPC stacks costs more than estimated	Land the broker side first; a client that cannot do TLS yet keeps the development flag and the start-up warning
Staleness handling changes the visual design	Degraded presentation is specified in PULSE-SAF-001 section 3; no new design work, only the agreed dashes, greying and banner
Safety and security managers not assigned in time	Item 14 slips to sprint 3 without blocking the gate; the documents stay marked "not reviewed"

6. Review feedback, 4 September

The review was held on 4 September as planned. No comments were raised and no requirement changed as a result, so this table is empty by outcome rather than by omission.

#	Comment	Raised by	Action	Target sprint
-	No comments raised at the review	-	None	-

7. Definition of done for the week 6 gate

1. Regression suite passes on a clean checkout and its report is committed under docs/evidence/.
2. The cluster shows degraded state within the freshness criterion when the broker is stopped, on both HMIs.
3. Broker and all four clients run over mutual TLS; a plaintext start prints the warning.
4. PULSE-SyRS-001, PULSE-SAF-001 and PULSE-SEC-001 status columns updated and PULSE-RTM-001 regenerated with zero dangling references.
5. Demo rehearsed on the strip display and timed under five minutes.

8. Progress at the close of week 4

Every state below was checked against the repository on 11 September, not self-reported. Where an item is partly done, the gap is named.

#	Item	Est. h	Target at the gate	State at week 4	Checked by
1	Regression suite	16	Implemented	Done, but runs by hand	Nine checks and six committed reports under docs/evidence/. No CI workflow exists, so nothing runs it automatically

#	Item	Est. h	Target at the gate	State at week 4	Checked by
2	Signal contract test	12	Implemented	Partly done	The check resolves 28 appendix A paths in the compiled tree. It does not subscribe and receive a typed datapoint, which is what the item asks for
3	UI smoke test per build	20	Implemented	Barely started	flutter-launch asserts the process stays up for four seconds. No injected value, no assertion on what is rendered, nothing on Compose
4	Flutter staleness display	32	Implemented	Not started	No freshness or staleness logic in hand-written HMI source
5	Compose staleness display	16	Implemented	Not started	As above
6	Mutual TLS on the broker	24	Implemented	Not started	scripts/start-databroker.sh still passes --insecure
7	Pin the Flutter SDK	6	Partial	Half done	Broker pinned by tag and digest. No FVM pin and no SDK version recorded in the README
8	SBOM and vulnerability check	14	Implemented	Not started	No SBOM file and no check
9	Driver-monitor privacy test	6	Implemented	Not started	No check. The claim holds by inspection but nothing proves it
10	HMI-side range check	10	Implemented	Not started	Broker-side validation only
11	Catalogue checksum	4	Implemented	Not started	No check
12	Merge the monitoring branch	4	Done	Done	Merged into main on 4 September
13	Architecture write-ups	16	Implemented	Not started	No SWAD documents exist
14	Safety and security review	8	Reviewed	Not held	Both concepts still carry “not safety-reviewed” and “not security-reviewed”
15	Virtual CAN feeder spike	8	Partial	Not started	No vcan0 or provider work. The only matches in the tree are diagram labels

8.1 Hours

Position	Hours
Committed in section 3	196
Delivered by the close of week 4	about 33
Remaining	about 163
Capacity left, weeks 5 and 6	160

The sprint reserve of 44 hours was held for review feedback and risk. The review raised no feedback, so that reserve was available. Week 4 spent it on work that was not in this backlog, so the position is now roughly break-even with no slack at all. One illness, one hard defect, or one more week like the last, and the gate is missed.

8.2 What week 4 actually produced

None of it was on this backlog, and all of it was worth doing. Said plainly so that it is not mistaken for progress against work package 02.

- Every download on the published documentation site returned 404. The page was built for one directory layout and published into another. All fourteen now serve, and each document is available as PDF or Word.
- The specification set was revised for readability, versioned and dated.
- A defect was found and closed: the odometer published kilometres into a signal the catalogue defines in metres, and the display divided by nothing, so the two errors cancelled and the dial looked correct. A new suite check, unit-sanity, fails the build if it recurs.

Against the backlog, week 4 moved item 1 by one check and item 7 by part of a decision. Nothing else.

8.3 Decisions requested

Each one is cheap this week and expensive in week 6.

#	Decision	Why it cannot wait
1	Cut item 5, Compose staleness	There is no Java, Gradle or Android SDK on the machine, so it would be written blind and untested. Cutting it is what returns slack to the sprint
2	Amend definition of done item 2	It says degraded state on both HMIs. If item 5 is cut, that wording must change before the gate, not at it
3	Fund parity or narrow the benchmark claim	FR-13 is Partial: Flutter subscribes to 28 signals, Compose to 17. The comparison in the deck assumes both consume the same data, and that is no longer true
4	Settle the Flutter SDK pin	Commit the 3.47 upgrade, or revert and pin through FVM. Six hours of work behind a decision open since sprint 1
5	Book the safety and security review	Item 14 needs two external calendars and there are two weeks left. It slips by default
6	Order the target hardware	Work package 03 gates at week 9. Section 7.2 of PULSE-SAD-001 specifies a Raspberry Pi 5 with 8 GB; if a 16 GB board is bought instead, that line changes

8.4 Proposed change to this plan

Add a checkpoint at the end of week 5, one hour, against this same table. A three-week sprint whose first status point is the gate itself has no room to recover. If week 5 goes the way week 4 did, that should be known on 18 September rather than on the morning of the demo.

8.5 State at the close of week 6

Checked against the repository on 23 September, two days before the gate. Nothing landed between the week 4 checkpoint and 23 September; everything below landed on 23 September.

#	Item	Target	State on 23 September	Checked by
1	Regression suite	Implemented	Implemented	16 checks, .github/workflows/regression.yml runs them on every push. Evidence under docs/evidence/
2	Signal contract test	Implemented	Implemented	live-contract: all 28 appendix A paths subscribe and deliver a typed datapoint from the running broker
3	UI smoke test per build	Implemented	Partial	Flutter: test/cluster_smoke_test.dart renders an injected 87 km/h in gear 4 and asserts it. Compose: unit-tested, no emulator without KVM on this host
4	Flutter staleness display	Implemented	Implemented	Degrades within 1 s of broker loss; screen captures and widget tests
5	Compose staleness display	Implemented	Partial	Same model in VssClient.kt, compiled and unit-tested; not yet seen on a device
6	Mutual TLS on the broker	Implemented	Implemented with deviation	TLS and per-client tokens, secure by default; transport-security proves the refusals. KUKSA 0.7.0 cannot verify client certificates (SEC section 5, DD-12)
7	Pin the Flutter SDK	Partial	Implemented	3.47.0 in .fvmlrc, floor in pubspec.yaml, toolchain-pin check
8	SBOM and vulnerability check	Implemented	Implemented	85 components, OSV clean after Pillow was fixed by moving to Python 3.11
9	Driver-monitor privacy test	Implemented	Partial	dms-privacy written; needs AVX to run the monitor, so skips here and runs in CI
10	HMI-side range check	Implemented	Implemented	Both HMIs read catalogue limits from the broker and refuse values outside them; range-refusal proves the broker refuses first
11	Catalogue checksum	Implemented	Implemented	catalogue-checksum; a one-character edit fails the run
12	Merge the monitoring branch	Done	Done	4 September
13	Architecture write-ups	Implemented	Implemented	PULSE-SWAD-001, one page each for the Flutter cluster and the providers
14	Safety and security review	Reviewed	Not held	Needs the two managers; nothing an engineer can do alone
15	Virtual CAN feeder spike	Partial	Implemented	scripts/can_spike.py: three signals from vcan0 through AGL's DBC and KUKSA's provider, over TLS, into the catalogue

Definition of done for the gate (section 7): 1 met; 2 met on Flutter, not yet shown on Compose; 3 met with the deviation above; 4 met; 5 the bring-up was timed at 3.9 s on a virtual 2560 x 720 screen, the physical strip is not on this host.

Decisions taken while implementing, for the review to confirm or reverse: Python 3.11 for the providers (the reason is in requirements.txt); Flutter stays at 3.47.0; freshness is judged by arrival time rather than the broker's timestamp (SAF section 3); token authentication stands in for mutual TLS (SEC section 5). One finding: the driver monitor cannot run on this VM because it exposes no AVX, so the sprint 3 target and the demo host must.

8.6 State on 2 October, and the week 7 checkpoint

The three items left partial at the gate needed hardware the reference VM does not have: a CPU with AVX, KVM for an Android emulator, and the strip display. A second host has all three. Checked against the repository on 2 October; every row below is in a commit of that day.

#	Item	State on 23 September	State on 2 October	Checked by
3	UI smoke test per build	Partial	Implemented	Compose: ClusterSmokeTest.kt on the Android 16 car emulator renders an injected 87 km/h in gear 4, then the degraded form; suite check compose-ui
5	Compose staleness display	Partial	Implemented	All eight criteria, as in Flutter; captures 1.4 s after the broker was stopped and after it returned (docs/evidence/screens/)
9	Driver-monitor privacy test	Partial	Implemented	dms-privacy ran with the monitor executing for the first time. It failed, correctly, and the cause was fixed (finding 1)
15	Virtual CAN feeder spike	Implemented	Implemented, extended	The simulator can act as the vehicle's ECUs on vcan0 (CAN=1 ./run.sh); can-decode covers all 35 mapped signals in CI

Definition of done for the week 6 gate (section 7), restated: 1 met, and CI now actually runs the suite (finding 3); 2 met on both HMIs; 3 met with the CS-3 deviation; 4 met; 5 met on the physical strip: ./run.sh to the cluster on the strip in 4.1 to 5.1 s with the broker start included, and the broker-loss demo recorded as live, degraded and recovered (docs/evidence/screens/20261002-flutter-strip-*).

FR-13 is Implemented: both clusters consume the same 28 signals. That answers decision 3 of section 8.3 by funding parity. The benchmark in the deck predates it and is re-measured in sprint 4.

Findings of the day:

- MediaPipe sends usage statistics to Google.** The driver monitor resolved play.googleapis.com and opened HTTPS connections to it. The library's own logger, present in 0.10.35 and 1.0.1 (the versions tested), has no off switch. No camera data is in those reports, but CS-6 allows no socket except the broker. The logger is now held on the host; PULSE-SEC-001 section 5 records the finding and why the fix is a workaround.
- Installs drifted.** Two transitive packages had moved since the SBOM was reviewed, so the sbom check failed. constraints.txt now pins the whole resolved set; a fresh install matches the SBOM exactly.

3. **CI had never run the suite.** Every run since 23 September stopped at a missing apt-get update. Fixed; CI also fetches VSS v6.0 so the regeneration check runs there.
4. **The Compose cluster could not authenticate on Android Automotive.** Apps run as user 10 there, and the credential script wrote to user 0. Fixed in scripts/android-push-pki.sh.
5. **The simulator did not survive a broker restart.** It now reconnects.
6. **The Flutter cluster did not recover from a broker restart.** It degraded correctly, but the Dart gRPC stream stayed open after the broker died and the client never resubscribed. Three seconds of silence now ends the stream; recovery measured at 2.6 to 3.9 s. Found only on the strip rehearsal, which is the argument for rehearsing on it.
7. **An empty entry counted as an arrival.** A new subscription opens with an empty entry for every path without a value; both clusters took it as fresh, so an absent driver monitor raised no warning for one second after reconnecting. Both now ignore empty entries; Flutter test added.
8. **The suite left driver monitors running.** Ending strace in dms-privacy did not end the traced monitor, and each orphan kept publishing driver state. The suite now ends whole process groups.
9. **The CAN provider forwarded the hazard only on change.** The catalogue maps it with on_change, so in CAN mode an unchanged hazard would go stale after 2 s. The provider's own mapping copy forwards every safety signal at 100 ms; can-path asserts the hazard stays fresh.

Week 7 checkpoint for sprint 3 (gate 16 October: CAN on target hardware):

Sprint 3 item	State on 2 October
Compose staleness shown on a device	Done (rows 3 and 5 above)
Safety and security review	Not held. Needs the two managers' calendars
Physical CAN bus behind the same contract	Virtual path complete; CAN HAT not started. The provider, DBC and checks do not change for can0
Raspberry Pi 5 baseline	Not started. No board recorded as ordered. It must expose the CPU features the driver monitor needs (finding of 23 September)

Decisions still open from the review: book the safety and security review; order the Pi 5 (8 GB as PULSE-SAD-001 section 7.2 specifies, or 16 GB with that line changed); change the wording of CS-3 or plan a broker with client-certificate support.